Calibration-Split Leakage in Selective-Prediction Retrieval-Augmented Generation: A Re-Validation Case Study with Confirmed Gains from Residual Vector Quantization and OCR-Confidence Calibration

Manthira Moorthy P, Vijay S K

Final Year, Department of Electronics and Communication Engineering, Mepco Schlenk Engineering College, Sivakasi
Mentor: Dr. V. Karthikeyan, Assistant Professor (Sl. Grade), Department of ECE, Mepco Schlenk Engineering College

Unpublished student project manuscript. Not peer-reviewed, not submitted to any journal or conference. Every numerical claim in this document is either quoted from a cited published source, or produced by this project's own code, logged under research/experiments/, and reproducible via the commands named in Section 4. This manuscript supersedes an earlier internal draft of the same system whose headline claim is shown in this paper to not survive a corrected evaluation methodology — that correction is itself this paper's primary reported finding, not a footnote.

Abstract

Selective-prediction mechanisms for retrieval-augmented generation (RAG) — decision layers that withhold or redirect an answer when retrieved evidence is judged insufficient — are typically validated by fitting decision thresholds on a calibration split and reporting accuracy gains on a held-out evaluation split. We show, via a controlled before/after case study on one such mechanism (Evidence-Adaptive Retrieval and Verification, EAR, a four-signal logistic sufficiency score added to an OCR-resilient hybrid-retrieval document-QA pipeline), that a split which is disjoint at the query level but not at the source-document level can silently inflate the apparent benefit past the point of statistical significance. On the OHR-Bench benchmark (Zhang et al., ICCV 2025), an initial evaluation with 51% document overlap between calibration and evaluation reported a statistically significant +7.1-point selective-accuracy improvement (71.2%→78.3%, 95% CI [+2.5, +12.0], 62.8% coverage). Under a corrected, verified-zero-overlap, document-disjoint split, the same mechanism's improvement trends positive but is not statistically significant (62.3%→68.4%, 95% CI [−8.3, +19.7], coverage collapsing to 15.4%). We report this corrected, weaker result as our primary finding rather than retaining the stronger, invalid one. Separately, and unaffected by this leakage mode because neither uses a calibration/evaluation split, we confirm two further contributions on the same pipeline: (1) a Residual Vector Quantization (RVQ) recall gap is root-caused to quantization reconstruction error (Welch's $t=-4.71$, $p=3.5\times10^{-6}$) and 79% recovered by a quantize-then-exact-rerank strategy at no added latency; (2) real per-word OCR confidence, captured at ingestion, is a statistically reliable proxy for actual character error rate under controlled degradation (Pearson $r=-0.984$, $p=1.4\times10^{-30}$), even though this specific signal's weight is empirically near-zero inside the combined EAR model in both the original and corrected fits — an independently confirmed case of a validated signal being functionally inert inside a jointly-fit model. We argue the leakage case study itself is this paper's most generalizable contribution: a concrete, reproducible demonstration of how document-level split construction, not just query-level disjointness, determines whether a selective-prediction claim in RAG is actually defensible. A threshold-independent re-analysis (Area Under the Risk-Coverage Curve = 0.287 against a full-coverage baseline risk of 0.377; Brier score 0.212 against a naive-predictor Brier of 0.235) shows EAR's sufficiency score remains informative even where its single fitted operating point is not significant, and a risk-coverage curve analysis identifies a lower-risk, nearby coverage region the fitted threshold does not reach — evidence favoring an underpowered-calibration explanation over a no-effect one, offered as a hypothesis for future recalibration rather than a second confirmed result.

This revision extends that re-analysis in two directions. First, a per-signal diagnostic (ROC-AUC, individually, for each of EAR's four raw signals, computed here for the first time on any split) finds the combined score's overall AUC is 0.674 and that two signals — agreement (AUC 0.665) and complexity (AUC 0.655) — are individually informative, while ocr_confidence is not merely weakly weighted but literally constant on this evaluation split (making a zero weight mathematically inevitable, not an empirical finding about the signal's value), and complexity is informative in the OPPOSITE direction from its design hypothesis. A post-hoc power analysis further shows this evaluation's sample size (n=38 answered) could only reliably detect an effect roughly 4× larger than the one observed (minimum-detectable Cohen's h = 0.488 vs. observed 0.128), reframing the non-significant result as expected under under-powering rather than as evidence of no effect. Second, this revision reports a production-engineering contribution, reported separately from and not as evidence for the statistical findings above: the EAR pipeline was implemented as a real, independently runnable service — a FastAPI backend with persistent document storage, a citation-verification pipeline that checks every

generated citation against the actual retrieved evidence, and deletion that is programmatically verified rather than assumed — and exercised end-to-end, including by driving a real web browser against the running system, not merely by inspecting source code.

Finally, we act on the diagnostics rather than only report them: EAR v2 drops the zero-variance ocr_confidence signal and replaces the logistic sufficiency score's linear probability mapping with an isotonic recalibration fit on the calibration split, both changes pre-specified, before results were inspected, to fix calibration quality specifically and NOT to be expected to resolve the power-limited significance gap. The result matches that prediction exactly: Expected Calibration Error nearly halves (0.096→0.046) while coverage, combined ROC-AUC, and statistical significance are essentially unchanged — a successful, falsifiable prediction about which problem a given fix can and cannot solve, which we argue is stronger evidence of a correct diagnosis than an unexplained improvement would have been.

Keywords: Retrieval-Augmented Generation, Selective Prediction, Calibration-Evaluation Leakage, Residual Vector Quantization, OCR-Resilient Document QA, Reproducibility, Evidence-Adaptive Verification, ROC-AUC, Calibration, Statistical Power, Production System Architecture, Isotonic Regression, Model Recalibration.

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1. Introduction

Retrieval-Augmented Generation (RAG) systems answer questions by retrieving passages from a corpus and conditioning a language model's output on them. In practice, most RAG pipelines — including QuIM-RAG (Saha, Saha & Malik, IEEE Access, 2024), the question-indexed retrieval baseline this project extends — retrieve some evidence and then unconditionally generate an answer, with no mechanism to recognize that retrieval itself may have failed. This is sharper for OCR-processed document collections, where recognition noise, broken table structure, and fixed-size chunking degrade evidence quality without any visible signal to the system or the end user.

A natural response, increasingly explored in 2025–2026 RAG literature, is a selective-prediction or abstention layer: compute a confidence signal from retrieval, and withhold or redirect the answer when confidence is low. This project implements one such layer, Evidence-Adaptive Retrieval and Verification (EAR), and initially reported — in an earlier internal draft of this same system — a statistically significant selective-accuracy improvement from it. This paper reports the outcome of a subsequent, deliberately adversarial re-validation of that claim: a systematic audit that treated no prior number as trusted, re-extracted the evaluation protocol, and found a document-level leakage defect in the calibration/evaluation split construction used throughout the earlier draft.

This paper's contribution is therefore not primarily "EAR works." It is a corrected, honestly reported re-measurement of EAR under a fixed evaluation protocol — which found the original claim does not hold — together with two further contributions (a quantization recall-loss fix, and an OCR-confidence calibration result) that are unaffected by the leakage mode and remain fully supported. We present the leakage finding itself as a methodological contribution with relevance beyond this specific system: any selective-prediction or abstention evaluation in RAG that shares a document corpus between its calibration and evaluation splits is exposed to the same risk.

1.1 Research Questions

- RQ1: Does restricting RAG answers to a calibrated sufficiency-score decision (EAR) measurably improve top-1 retrieval accuracy, and does that improvement survive a genuinely document-disjoint calibration/evaluation split, not merely a query-row-disjoint one?
- RQ2: What is the quantitative effect of document-level calibration/evaluation leakage on a selective-prediction claim's reported effect size and statistical significance in this setting?
- RQ3: Is a residual-vector-quantization-induced retrieval-recall gap attributable to a specific, testable mechanism (quantization reconstruction error), and can a low-cost fix recover the lost recall without added latency?
- RQ4: Is real per-word OCR confidence, captured at ingestion, a statistically reliable proxy for actual character error rate under controlled degradation, independent of whether it is useful inside a specific downstream decision model?
- RQ5: Independent of the combined model's coverage/accuracy tradeoff, does each of EAR's four raw signals individually carry information about retrieval correctness, in the direction its design hypothesized — and is the evaluation that tests RQ1 statistically capable of detecting an effect of the size actually observed?
- RQ6: Can the EAR pipeline be implemented as a real, independently runnable, multi-component system (API, persistence, citation verification) and have that implementation verified by actually exercising it end to end, rather than only by inspecting its source code?

- RQ7: Do fixes derived directly from RQ5's diagnostics (dropping an uninformative signal; recalibrating a non-monotonic probability-to-accuracy mapping) measurably improve EAR's calibration quality, and does doing so also resolve RQ1's statistical significance gap, as a naive reading might expect?

1.2 Contributions

- A controlled, reproducible before/after case study demonstrating that document-level (not merely query-level) calibration/evaluation disjointness materially changes a RAG selective-prediction claim's statistical significance — from a 95% CI that excludes zero to one that includes it — on identical underlying data and code, differing only in split construction (RQ1, RQ2).
- A statistically confirmed root cause (Welch's t-test) for a Residual Vector Quantization recall gap, and a quantize-then-exact-rerank fix recovering 79% of the lost Recall@5 at no measurable added latency (RQ3).
- A confirmed, strong correlation between real Tesseract OCR confidence and actual character error rate under controlled synthetic degradation, reported alongside the finding that this signal is nonetheless functionally inert inside the specific combined decision model tested — reported as a calibration-design finding, not a signal-quality failure (RQ4).
- A threshold-independent re-analysis of EAR's own sufficiency score via risk-coverage curves, Area Under the Risk-Coverage Curve (AURC), and Brier-score calibration, showing the score itself carries real information even though its specific fitted operating point does not reach significance — together with Wilson-interval and McNemar's-test cross-checks against the bootstrap-based tests used elsewhere in the paper, which surface a genuine, explained disagreement between two valid significance tests applied to the same retrieval comparison.
- The first per-signal ROC-AUC analysis of EAR's four raw signals on any split, finding two (agreement, complexity) individually informative and two (ocr_confidence, contradiction) not — including the discovery that ocr_confidence is literally constant, not merely weakly weighted, on the corrected evaluation split — plus a post-hoc power analysis showing this evaluation's sample size could only reliably detect an effect roughly 4x larger than the one observed, quantifying rather than merely asserting the "underpowered, not necessarily absent" reading of RQ1's result (RQ5).
- A working, independently verified production implementation of the full EAR pipeline — a FastAPI backend with persistent document storage, retrieval-time citation verification, and programmatically-checked deletion — exercised end to end via both direct API calls and a real browser session, reported as an engineering contribution distinct from, and not offered as evidence for, the statistical findings above (RQ6).
- EAR v2: a diagnosis-driven redesign (dropped zero-variance signal, isotonic recalibration) that, exactly as pre-specified before its results were inspected, nearly halves calibration error (ECE 0.096→0.046) without changing coverage, combined ROC-AUC, or statistical significance — a successful, falsified-in-advance prediction that separates and independently resolves the calibration-quality problem from the statistical-power problem the power analysis (RQ5) identified as the actual constraint on RQ1 (RQ7).

2. Related Work

QuIM-RAG (Saha, Saha & Malik, 2024) is the direct baseline: an inverted question-index retrieval mechanism evaluated on a private NDSU catalog/career-advising corpus, reporting large Faithfulness

and BERTScore gains over a direct chunk-cosine baseline. Its evaluation has no OCR noise, no lexical retrieval path, no reranking, and — most relevant here — no calibration/evaluation split of any kind, since it has no selective-prediction mechanism and reports single point estimates with no significance testing (Table 2 of the original paper).

Selective prediction and abstention in RAG is an active area: verification-model-gated answering, evidence-sufficiency abstention benchmarks, confidence-adaptive retrieval budgets, and NLI-style contradiction detection between retrieved passages are all represented in the recent literature our system's design draws on. None of this literature's abstention-evaluation protocols that we reviewed explicitly distinguishes query-level from document-level disjointness when constructing a calibration/evaluation split — which is precisely the distinction this paper's central finding turns on.

OHR-Bench (Zhang et al., arXiv:2412.02592, ICCV 2025) directly motivates this project's OCR-robustness framing and supplies the public, human-verified, page-localized benchmark used throughout this paper, since QuIM-RAG's own NDSU corpus is private and unavailable. Product Quantization (Jégou, Douze & Schmid, 2011) is the foundation of this project's Residual Vector Quantization module, which implements a simplified two-stage instance of it without the per-subspace splitting production approximate-nearest-neighbor systems typically add. BM25 (Robertson & Zaragoza, 2009) is the lexical scoring function fused with dense retrieval via Reciprocal Rank Fusion in this project's hybrid retrieval path.

On leakage specifically: train/test contamination at the entity or document level is a well-known risk in general machine learning evaluation, but we are not aware of prior published work that explicitly quantifies its effect on a RAG selective-prediction claim's statistical significance in the way Section 5 of this paper does — this specificity is what we consider this paper's most transferable contribution.

3. Proposed Methodology

This section describes (a) the system under study, EAR, exactly as implemented and unchanged by this paper's re-validation; and (b) the corrected split-construction methodology this paper introduces, which is the actual object of Section 5's central experiment.

3.1 System architecture

```
PDF / scanned document
  -> PyMuPDF text extraction, else Tesseract OCR (+ real per-word confidence capture)
  -> paragraph-aware chunking (950 characters, 100-character overlap)
  -> MiniLM embeddings (384-dim; TF-IDF fallback) + real BM25 index, built side by side
  -> retrieval: Dense cosine | BM25 | Hybrid (Reciprocal Rank Fusion, k=60)
  -> EAR: compute 4 signals -> sufficiency score p -> decision
      - answer             -> generate + groundedness pre-check
      - expand_retrieval   -> re-retrieve with larger k, then answer
      - inspect_structure  -> surface conflicting evidence for human review
      - abstain            -> no answer generated
```

A block diagram reproducing this flow (see Supplementary Materials) is drawn directly from quimrag/core.py, bm25.py, adaptive.py, and rvq.py.

3.2 EAR signal formulation (unchanged from the system under study)

Retrieval agreement — Jaccard overlap of top-K chunk-id sets from Dense-only and BM25-only rankings, both already computed for Hybrid RRF:

$$a = |D_K \cap L_K| / |D_K \cup L_K|$$

(1)

OCR confidence — mean real Tesseract per-word confidence across the top-K evidence chunks:

$$o = (1/K) \cdot \Sigma_{c \in E_K} conf(c)$$

(2)

Cross-passage contradiction — fraction of topically-related evidence pairs with disjoint extracted number sets:

$$x = |\{(i,j) : overlap(i,j) > 0.3 \wedge nums(i) \cap nums(j) = \emptyset\}| / C(K,2)$$

(3)

Query complexity — a bounded proxy from token count, wh-word count, and conjunction count:

$$c = mean(min(tok,30)/30, min(wh,3)/3, min(conj,3)/3)$$

(4)

Sufficiency score — logistic combination, weights fit by logistic regression on the calibration split only:

$$p = \sigma(w_0 + w_1 a + w_2 o + w_3 x + w_4 c)$$

(5)

Decision thresholds θ_abstain, θ_answer are also fit on the calibration split: θ_answer is set to the lowest calibration probability at which precision for predicting top-1 correctness reaches ≥0.8 (falling back to the best-F1 threshold if 0.8 precision is unreachable on that calibration set), and θ_abstain is the 20th percentile of calibration probabilities — both fit, not hand-picked. Pseudocode (quimrag/adaptive.py, unmodified by this paper):

```
def decide(p, signal, theta_abstain, theta_answer):
  if p < theta_abstain: return "abstain"
  if p < theta_answer:
    weakest = argmax_f badness(f, signal[f])
    return "inspect_structure" if weakest == "contradiction" else "expand_retrieval"
  return "answer"
```

3.3 The methodological contribution: document-level split construction

The system's pre-existing calibration/evaluation split (used in the earlier draft) partitioned individual query rows uniformly at random, verifying only that no exact query row appeared in both pools. Because each source document in OHR-Bench contributes multiple query rows, this leaves a document free to have some of its queries in calibration and others in evaluation — a document-level leak, even though the split is query-row-disjoint by construction.

This paper's corrected procedure instead partitions by document identity first, then assigns a document's entire query set as a unit:

```
def split_by_document(qas, calib_frac, seed):
    docs = group_by(qas, key=doc_name)
    shuffle(docs, seed)
    calib, evaluation = [], []
    running = 0
    for doc_name, rows in docs:
        target = calib_frac * total_queries
        (calib if running < target else evaluation).extend(rows)
        running += len(rows)
    assert doc_ids(calib) & doc_ids(evaluation) == {}   # verified, not assumed
    return calib, evaluation
```

The zero-overlap property is verified programmatically after every split construction (Section 4.4) and is additionally pinned by a dedicated regression test (tests/test_ohrbench_adapter_v2.py) added as part of this paper's own validation work, so this specific defect class cannot silently reappear in future experiments.

3.4 Residual Vector Quantization and its fix (unchanged from the system under study)

Two-stage k-means residual quantization: stage 1 assigns x to its nearest of N centroids $C^{(1)}$; stage 2 quantizes the residual against a second codebook $C^{(2)}$:

$$i_1 = \mathrm{argmin}_j \|x - C^{(1)}_j\|, \quad r = x - C^{(1)}_{i_1}, \quad i_2 = \mathrm{argmin}_j \|r - C^{(2)}_j\|, \quad \hat{x} = C^{(1)}_{i_1} + C^{(2)}_{i_2} \tag{6}$$

The candidate fix tested in Section 5.3 narrows the quantized search to a small candidate pool, then reranks that pool using exact (unquantized) vectors — trading a kept exact-vector buffer for recovered recall at the cost of its own storage footprint, a tradeoff this paper's experiment measures for latency and recall but not for storage (see Section 7).

3.5 Evaluating EAR Beyond a Single Threshold: Selective-Prediction Diagnostics

Section 5.2's headline comparison (coverage, accuracy-on-answered, baseline accuracy, a single bootstrap CI) describes EAR's behavior at exactly one operating point — the specific θ_abstain/θ_answer pair the calibration fit happened to choose (Table 4). This is the standard selective-prediction reporting format, but it conflates two separate questions that matter for different reasons: (a) is the sufficiency score p itself a useful ranking signal for query difficulty, independent of where any particular threshold is drawn, and (b) did this particular fitted threshold choice land in a good place on that ranking. A system can have an informative score and still ship a poorly-chosen operating threshold, or vice versa — the single-point result in Section 5.2 cannot distinguish these. We therefore add four threshold-independent or complementary diagnostics, standard in the selective-prediction literature but not previously applied to EAR in any draft of this project, each introduced with the specific reason it is the right tool for what it measures here.

3.5.1 Risk-coverage curve and Area Under the Risk-Coverage Curve (AURC)

Following the selective classification framework of El-Yaniv and Wiener (2010) and Geifman and El-Yaniv (2017): for a classifier with confidence score p and a rejection threshold θ, define coverage and selective risk as functions of θ over the evaluation set of size n:

$$coverage(\theta) = (1/n)\,\Sigma_i\,\mathbb{1}[p_i \geq \theta]$$

(7)

$$risk(\theta) = 1 - (\,\Sigma_i\,\mathbb{1}[p_i \geq \theta]\cdot correct_i\,) / (\,\Sigma_i\,\mathbb{1}[p_i \geq \theta]\,)$$

(8)

Sweeping θ over [0,1] traces a risk-coverage curve; its area, AURC, summarizes selective-prediction quality as a single number independent of any one threshold choice:

$$AURC = \int_0^1 risk(coverage)\,d(coverage)$$

(9)

We use this specifically because Section 5.2's single-point result cannot tell us whether EAR's p is a weak ranking signal overall, or a reasonable ranking signal paired with a threshold that the small calibration sample (n=173) happened to place suboptimally. AURC answers the first question directly, independent of that threshold-selection step; Section 5.6 reports both the curve and the fitted operating point's position on it, so a reader can see whether the fitted threshold is actually near the curve's best region.

3.5.2 Brier score: is p itself well-calibrated?

Independent of the decision thresholds, p is meant to behave as a probability — larger p should mean a genuinely higher chance the top-1 retrieved chunk is correct. The Brier score, a strictly proper scoring rule, measures exactly this:

$$Brier = (1/n)\,\Sigma_i\,(p_i - correct_i)^2$$

(10)

We use Brier score rather than log-loss specifically because log-loss diverges to infinity whenever a confidently-wrong prediction occurs ($p \to 0$ with $correct_i=1$, or $p \to 1$ with $correct_i=0$), which would let a handful of overconfident mistakes dominate the metric on a sample as small as n=247; Brier score stays bounded in [0,1] and degrades gracefully, which better suits a small-sample diagnostic. As a reference point we also report the Brier score of a naive predictor that always outputs the dataset's overall base rate of correctness (ignoring p entirely) — EAR's p beating this naive baseline is a necessary (though not sufficient) condition for p to be carrying real information, separate from whether the specific fitted threshold produces a significant accuracy gain.

3.5.3 Wilson score interval: a closed-form cross-check on coverage

Every CI elsewhere in this paper is a bootstrap CI. For a single proportion — EAR's coverage, k "answer" decisions out of n evaluation queries — a closed-form alternative exists and is worth reporting as an independent cross-check, since it does not share the bootstrap's resampling-based construction:

$$CI_Wilson = (\,\hat{p} + z^2/2n \pm z\sqrt{(\hat{p}(1-\hat{p})/n + z^2/4n^2)}\,) / (1 + z^2/n),\;\; \hat{p} = k/n$$

(11)

We use the Wilson interval rather than the simpler normal (Wald) approximation $\hat{p} \pm z\sqrt{(\hat{p}(1-\hat{p})/n)}$ specifically because the Wald interval is known to perform poorly — sometimes producing bounds outside [0,1], or being badly mis-calibrated — exactly in the small-n, extreme-$\hat{p}$ regime this paper's corrected coverage falls into (k=38, n=247, $\hat{p}$≈0.154); Wilson's interval remains well-behaved there by construction.

3.5.4 McNemar's test: a paired alternative for the baseline comparison

Section 5.1's Dense-vs-Hybrid comparison uses a paired bootstrap on nDCG@5, which is appropriate for a continuous, graded ranking metric. Because Dense-only and Hybrid-RRF are evaluated on the identical set of 247 queries, a second, classical paired test is also directly available and worth cross-checking against: McNemar's test on the 2×2 table of (Dense top-1 correct, Hybrid top-1 correct) outcomes per query, which asks a narrower, binary question — does Hybrid's ranking actually change which specific queries get the single best chunk right, independent of how the rest of the top-5 ranking looks:

$$\chi^2 = (|b - c| - 1)^2 / (b + c), \quad b = n(Dense✓, Hybrid✗), \quad c = n(Dense✗, Hybrid✓)$$

(12)

with continuity correction (used because the discordant-pair count b+c, while not tiny, is small enough that the uncorrected chi-square approximation would overstate significance). We report both tests side by side in Section 5.1b deliberately: they measure different things (ranking quality across the full top-5 vs. narrow top-1 correctness) and are not required to agree — a disagreement is itself informative about which part of the ranking actually changed.

3.5.5 Cohen's h: separating statistical from practical significance

A confidence interval answers "is this detectable given the sample size"; it does not answer "how large is this, in a sample-size-independent sense." For two proportions p_1, p_2 (e.g., accuracy-on-answered vs. baseline accuracy), Cohen's h applies the arcsine-square-root variance-stabilizing transform so the resulting effect size is comparable across different sample sizes and different base rates:

$$h = 2 \cdot arcsin(\sqrt{p_1}) - 2 \cdot arcsin(\sqrt{p_2})$$

(13)

Conventionally |h|≈0.2 is a small effect, ≈0.5 medium, ≈0.8 large (Cohen, 1988). We compute h for the accuracy-on-answered-vs-baseline comparison specifically because Section 5.2's non-significant CI could otherwise be misread as "there is definitely no effect" rather than the more precise "this sample cannot distinguish a small effect from no effect" — h quantifies which of these the point estimate actually corresponds to, reported in Section 5.2b.

3.5.6 Per-signal ROC-AUC: which signal, if any, is actually doing the work?

Every prior document in this project's own validation trail — including the leave-one-signal-out ablation discussed in Section 5.4 and this paper's earlier drafts — identified EAR's four signals only through their fitted logistic weights inside the COMBINED model. A weight near zero (ocr_confidence) is consistent with either "this signal carries no information" or "this signal is informative but redundant with the others once they are jointly fit" — the two have different implications and the combined-model weight alone cannot distinguish them. The standard way to

test a single signal's OWN discriminative power, independent of the others, is to score it alone as a binary classifier against the ground-truth label (top-1 retrieval correctness) via its ROC-AUC:

$$AUC(f) = P(\, f(x_i) > f(x_j) \mid correct_i = 1,\ correct_j = 0 \,)$$

(14)

i.e. the probability that signal f ranks a randomly chosen correctly-answered query above a randomly chosen incorrectly-answered one. AUC=0.5 is chance; AUC>0.5 (or <0.5, with the sign reversed) indicates real discriminative power. This had never been computed for any individual EAR signal, on any split, before this analysis — it is implemented and run for the first time here against the corrected, document-disjoint evaluation split (Section 5.2d).

3.5.7 Expected Calibration Error and the reliability diagram

The Brier score (Section 3.5.2) is a single scalar and can mask qualitatively different failure modes — a model that is uniformly slightly overconfident and one that is overconfident in one probability range and underconfident in another can share a Brier score. Expected Calibration Error makes this visible by binning predictions and comparing each bin's mean predicted probability to its observed accuracy (Guo et al., 2017):

$$ECE = \Sigma_\{bin\ b\}\ (n_\beta/n) \cdot |mean_predicted_p(b) - observed_accuracy(b)|$$

(15)

Bins are constructed by equal frequency (quantiles of p), not equal width, because EAR's probabilities are concentrated in a narrow range on this evaluation split (Section 5.2d) — fixed-width bins would leave several bins with too few queries for a stable accuracy estimate, which equal-frequency binning avoids by construction.

3.5.8 Post-hoc power analysis: was this evaluation even capable of detecting the observed effect?

Section 5.2's non-significant result (95% CI [−8.3, +19.7] including zero) is consistent with two different underlying realities: the true effect is near zero, or the true effect is real but this specific sample (n=38 answered, n=247 overall) was simply too small to detect it reliably. A post-hoc minimum-detectable-effect (MDE) analysis, using the same arcsine-stabilized transform already introduced for Cohen's h (Section 3.5.5), distinguishes these by computing the smallest effect size this sample size could have detected at a conventional 80% power:

$$h_min = (z_\{\alpha/2\} + z_\{power\}) \cdot \sqrt{(1/n_1 + 1/n_2)}$$

(16)

with α=0.05 (two-sided, matching the CI elsewhere in this paper) and power=0.80 (the conventional target). This is reported as an approximation — the actual test used for Section 5.2's headline result is a paired bootstrap on a selected subset, not an independent two-proportion z-test, so h_min here is an order-of-magnitude diagnostic, not a replacement for the bootstrap CI — but it directly answers the question Section 6's discussion raises without, until this analysis, being able to quantify.

3.6 System Architecture: From Research Prototype to Production Implementation

Everything above describes EAR as a research question. This section describes the second contribution of this work: turning the research prototype (a single-process Streamlit application with no HTTP interface, no persistent document store, and no deletion mechanism) into a real, independently runnable service, and verifying — by actually running it, not by inspecting the code — that the resulting system behaves correctly end to end.

3.6.1 Why a modular monolith, not microservices

The new backend is a single FastAPI process that imports the existing quimrag/ library directly, rather than a set of independently-deployed services communicating over a network. This is a deliberate choice, not a shortcut: the target deployment is a single machine (Section 4.3's hardware), there is no measured concurrent-load requirement that would justify the operational cost of service-to-service network calls, and the existing QuIMRAG class already has no framework-specific coupling, making it a clean library boundary to wrap without modification. One additive method, remove_document(), was added to support verified deletion (Section 3.6.3) — the only change made to the core research library for this entire architectural layer.

```
Browser / frontend (static HTML+JS)
  -> HTTP (CORS-restricted to localhost in the default, unauthenticated local-mode config)
FastAPI backend (single process)
  |- documents router  -> validate (extension/magic-bytes/size) -> sha256 hash -> dedup check
  |                    -> background task -> quimrag.core.extract_pdf/chunk/embed/BM25-index
  |                    -> SQLite: Document row (status: processing -> ready/failed)
  |- ask router        -> quimrag.adaptive.retrieve_adaptive (Hybrid retrieval -> EAR decision)
  |                    -> generation provider (local Ollama / extractive / Anthropic, explicit choice)
  |                    -> citation verification against the exact evidence set used
  |                    -> SQLite: QueryRecord (decision, signals, latency -- a routing audit trail)
  |- system router     -> live engine/device/model status (no cached or hardcoded numbers)
  (shared, lock-guarded QuIMRAG engine instance -- the same class used by the Streamlit research console)
```

This diagram reflects backend/ exactly as implemented, not a target design -- every arrow corresponds to a code path exercised in Section 3.6.5's verification.

3.6.2 Citation verification as a correctness mechanism, not a cosmetic feature

A generated answer is only as trustworthy as its citations are real. Every answer the backend returns is checked, programmatically, against the exact evidence set the generator was given: each [C_n] marker in the output text is extracted and compared against the chunk_id set actually retrieved for that query. An answer whose citations do not resolve to real, retrieved evidence is flagged (citations_verified=false) rather than silently trusted. This is distinct from, and complementary to, EAR's own sufficiency routing: EAR decides whether to generate an answer at all; citation verification checks whether the specific citations in an answer that WAS generated are grounded in what was actually retrieved, catching a generation-time failure mode (a fabricated or misattributed citation) that EAR's retrieval-time signals cannot see.

3.6.3 Deletion with verification, not deletion by assumption

Deleting a document must remove every chunk, embedding, and BM25 posting derived from it — not just a database row referencing it. The backend's delete endpoint removes the document from the live retrieval engine (clearing its chunks and triggering a full reindex of the embedding matrix and BM25 index), then independently re-queries the live engine's chunk list to confirm no chunk belonging to that document remains, returning that confirmation (verified_not_retrievable) as part

of the API response rather than assuming success from the delete call completing without an exception.

3.6.4 Measured, not assumed, hardware acceleration

The embedding and reranking models already preferred Apple Silicon's MPS backend over CPU in the pre-existing code, but this had never been benchmarked against CPU before this work. A controlled benchmark (5 repeats, 2 warm-up iterations, batch sizes 8/32/128) found MPS throughput at batch=128 of 1,446.9 documents/second versus CPU's 614.0 documents/second on the same hardware — a measured 2.36× speedup, not an assumed one. No NVIDIA/CUDA acceleration is claimed or possible on this hardware (no discrete GPU present), stated explicitly rather than left ambiguous.

3.6.5 Verification methodology: running the system, not just reading the code

The full upload → background ingestion → ask → citation verification → delete loop was exercised twice, by two different methods: first via direct HTTP requests (confirming validation rejects a spoofed-extension upload, duplicate content returns the same document without re-ingesting, and a deleted document is genuinely unretrievable), and second by driving a real web browser against the running backend — a file injected into an actual file-input element, the Upload/Ask/Delete buttons clicked as a user would, with each resulting state change (processing → ready, a generated answer with its citation verified, the document disappearing after deletion) confirmed by screenshot. Code that has not been run is not treated as verified anywhere in this paper.

3.7 EAR v2: A Diagnosis-Driven Redesign

Section 5.2d's diagnostics do not just describe EAR's weaknesses — they specify two concrete, testable fixes, each targeting a SPECIFIC, already-measured failure mode rather than a general restatement of "calibrate better." We implement both and evaluate the result (Section 5.6) with the same rigor (document-disjoint fitting, bootstrap CIs, no evaluation-split leakage) as every other result in this paper. This is EAR v2: not a new hypothesis invented separately from the evidence, but the direct algorithmic consequence of what Section 5.2d actually found.

3.7.1 Fix 1 — drop the signal with zero variance

Section 5.2d found ocr_confidence constant across all 247 evaluation queries, which makes its near-zero fitted weight mathematically inevitable rather than evidence of the signal's true value elsewhere. EAR v2 fits on the remaining three signals only (agreement, contradiction, complexity) — one fewer free parameter for a 173-row calibration fit to overfit, and one fewer signal computed at inference time, at zero cost to the information available to the model on this benchmark.

3.7.2 Fix 2 — isotonic recalibration for a non-monotonic reliability curve

Section 5.2d's reliability diagram (Fig. 5) is non-monotonic: the highest-probability bin is LESS reliable than the two bins below it. A linear logistic model cannot represent a non-monotonic probability-to-accuracy relationship by construction — whatever the true relationship between the raw score and actual correctness is, forcing it through a sigmoid of a linear combination cannot fix this. Isotonic regression (Zadrozny & Elkan, 2002) instead learns an arbitrary, non-decreasing step function mapping raw score to calibrated probability:

$$p_calibrated = g(p_raw),\ \ g\ monotonic\ non\text{-}decreasing,\ \ g = argmin_g\ \Sigma_i\ (g(p_raw_i) - correct_i)^2$$

g is fit by pool-adjacent-violators on the calibration split's (raw score, correctness) pairs only, then applied, frozen, to the evaluation split's raw scores — no evaluation-split label is used to fit g, preserving the same calibration/evaluation separation this paper's central finding established is essential. g can represent exactly the non-monotonic correction the logistic model cannot: if raw scores in a high range are empirically less reliable than scores just below them, g can map both to the same, lower calibrated probability, which a linear model is structurally unable to do.

3.7.3 What this fix is, and is not, expected to solve

Stated before Section 5.6's results are reported, consistent with this paper's frozen-protocol discipline elsewhere: both fixes target CALIBRATION QUALITY — how well the probability p reflects true correctness likelihood. Neither fix changes the amount of information available to the model (isotonic regression is a monotonic transform of the same raw score; dropping a zero-variance feature removes no information) or the calibration sample size (still n=173). The power analysis (Section 5.2d) identified sample size, not calibration quality, as the binding constraint on statistical significance — so EAR v2 is explicitly NOT expected to flip Section 5.2's non-significant result into a significant one. What it is expected to improve, if the diagnosis is correct, is the reliability of the probability estimate itself (lower ECE/Brier), independent of where any particular decision threshold is drawn — a real, separately useful property for any downstream use of EAR's score (e.g., ranking queries by confidence for human review) that does not depend on the threshold-based coverage/accuracy numbers reaching significance.

4. Experimental Setup

4.1 Dataset

QuIM-RAG's own evaluation corpus (22,609 GPT-3.5-generated questions over a scraped NDSU university-catalog and career-advising corpus) is private with no public release, no code repository, and no stated data license — confirmed unreachable, not merely assumed, during this project's audit of the original paper (Section 2). A substitute public benchmark is therefore required for any retrieval- or selective-prediction-level claim in this paper, and the choice of substitute is not incidental: it determines what can and cannot be tested about EAR's four signals. We selected OHR-Bench (Zhang et al., arXiv:2412.02592, ICCV 2025) over four other candidates for reasons specific to what each of EAR's signals needs to be exercised meaningfully, not merely because it is a convenient, well-known benchmark.

4.1.1 What OHR-Bench actually contains

OHR-Bench ships 8,500+ unstructured PDF pages across 7 domains (Textbook, Law, Finance, Newspaper, Manual, Academic, Administration), each page paired with human-verified ground-truth structured text, and 8,498 question-answer pairs (file qas_v2.json) drawn from 1,117 unique source documents. Each QA record carries an evidence_source field tagging which of five OCR-difficulty categories the question's supporting evidence comes from — plain text (3,409 records, 40%), table (2,275, 27%), formula (1,223, 14%), chart (765, 9%), or reading-order-dependent content (691, 8%) — and an answer_form field, of which 2,980 records (35%) are Numeric. This paper uses six of the seven domains (law, finance, textbook, manual, academic, news), substituting "academic" for the original sampling plan's "paper" label after confirming qas_v2.json's own taxonomy has zero records under "paper" (Section 3.3); "administration" is available but not used, to keep the domain-set size comparable to prior work's six-domain design.

4.1.2 Why this schema matters for EAR specifically, not just as a generic QA benchmark

- The contradiction signal (Eq. 4) extracts and compares numbers across evidence chunks — it has essentially nothing to detect on a benchmark without numeric content. OHR-Bench's 2,980 Numeric-answer records and 2,275 table-evidence records (table rows are a dense source of the kind of near-duplicate, easily-transposed figures this signal targets) give it real material to work with; a benchmark like SQuAD, built from prose Wikipedia passages, would not.
- The ocr_confidence signal (Eq. 2) requires documents where OCR actually degrades — table, formula, and chart content (2,275+1,223+765 = 4,263 records, 50% of the benchmark) is exactly where document parsing and OCR historically fail hardest, which is the specific reason OHR-Bench's own authors built it around this evidence_source taxonomy in the first place, rather than sampling uniformly from clean prose.
- The agreement signal (Eq. 1) benefits from queries with genuine lexical/semantic retrieval tension — OHR-Bench's domain variety (legal clause language vs. conversational newspaper prose vs. terse manual instructions) produces a wider range of BM25-vs-dense agreement behavior than a single-domain benchmark would, which is necessary for the calibration split to observe enough variation in this signal to fit a non-degenerate weight on it.
- Crucially, OHR-Bench additionally ships genuine OCR-perturbed variants of its own ground truth — formatting_noise and semantic_noise sets, each at mild/moderate/severe severity, generated by three real OCR/VLM parsing systems (GOT, MinerU, Qwen2.5-VL-72B) rather than synthetic image corruption. This paper's OCR-robustness study (Section 5.4) does NOT yet use these native perturbed variants — it instead renders clean ground-truth text to images and applies synthetic corruption (blur, resolution loss, skew, JPEG compression, noise, contrast reduction) through the project's own OCR pipeline. This is stated explicitly as a limitation (Section 7): OHR-Bench's own real-OCR-engine-perturbed data would be a strictly stronger test of the ocr_confidence signal than this paper's synthetic degradation, and switching to it is a concrete, scoped, low-effort follow-up — the perturbed files already exist in the benchmark distribution and require no new data collection.

4.1.3 Why not the alternatives

Table 1. *Benchmark selection, evaluated against what EAR's four specific signals need to be meaningfully exercised — not a generic leaderboard comparison.*

Benchmark	OCR / scanned content	Numeric/table evidence	Multi-domain	Page-level grounding	Verdict
OHR-Bench (used)	Yes — native perturbed variants at 3 severities	Yes — 41% table/formula	Yes — 6-7 domains	Yes — human-verified per page	Selected
BEIR	No	Varies by sub-dataset, not designed for it	Yes, but not OCR-relevant	No, passage-level only	Rejected: no OCR dimension at all, this project's central differentiator
Natural Questions	No — clean Wikipedia	Minimal	No — single domain	No	Rejected: no OCR, no table/numeric density, weak agreement-signal variety
HotpotQA	No — clean Wikipedia	Minimal	No	No	Rejected for this pass, but

					flagged as the strongest candidate specifically for stress-testing the complexity signal's multi-hop design in future work
MS MARCO	No	No	No — web passages	No	Rejected: passage-ranking focus, no document-QA or OCR angle
SQuAD	No — clean Wikipedia	No	No	No	Rejected: no numeric/table density for the contradiction signal, no OCR dimension

No benchmark in this comparison matches QuIM-RAG's original NDSU task (institutional FAQ retrieval over clean, non-OCR web text) closely — this is a deliberate, stated substitution rather than a like-for-like replication, chosen because it lets this paper test claims (OCR robustness, multi-domain generalization, numeric contradiction detection) the original paper's own dataset structurally cannot test at all, since it has zero OCR content and a single institutional domain.

4.1.4 The dataset-provenance correction itself

This paper uses qas_v2.json (8,498 records) throughout. Every experiment prior to this paper's own audit used an older file, qas.json (4,598 records, from the same benchmark release but not the file OHR-Bench's current README names as canonical). Both files share an identical schema and ground-truth construction method, so results built on qas.json are not void, but the correction is made explicit here rather than silently carried forward, since it is one half of the confound discussed in Section 5.2 and Section 7.

4.2 Splits

A stratified sample (~70 queries/domain target, seed 42) is partitioned by document (Section 3.3), yielding 173 calibration queries across 109 documents and 247 evaluation queries across 153 documents, with programmatically verified zero document overlap and zero gold-(document,page) overlap. A second, smaller, independently-sampled 70-query substitute set (seed 17) and a 60-query RAGAS-generation sample (seed 23) exist from the earlier draft and are reported in Section 5.4 with an explicit caveat that their own split construction has not yet been checked for the same document-overlap property (Section 7).

4.3 Hyperparameters and software

Retrieval depth k=5; chunking 950 characters / 100-character overlap; embeddings all-MiniLM-L6-v2 (384-dim); BM25 via rank_bm25.BM25Okapi with default k_1/b; RVQ 256 centroids, 2 stages. Python 3.11.9, macOS-26.4.1-arm64 (Apple Silicon, CPU/MPS, not a portability claim). HF_HUB_OFFLINE=1 / TRANSFORMERS_OFFLINE=1 set for all runs (required to avoid a network-hang defect found and fixed during this paper's validation work — Supplementary Materials).

4.4 Evaluation methodology

Baseline retrieval comparison: Recall@5, nDCG@5, MRR@5, latency; paired bootstrap p-value (2,000–5,000 resamples) vs. Dense-only, Bonferroni-corrected across the resulting comparison family (α=0.025 for 2 tests). EAR selective-prediction: logistic weights and θ_abstain/θ_answer fit exclusively on the calibration split; 95% CI on (accuracy-on-answered − unfiltered-baseline-accuracy) via 5,000-resample bootstrap, seed 42. Welch's two-sample t-test for the RVQ root-cause analysis; Pearson/Spearman correlation for the OCR-confidence validation.

Reproduction commands:

```
HF_HUB_OFFLINE=1 TRANSFORMERS_OFFLINE=1 .venv/bin/python research/run_benchmark_v2.py   # Section 5.1-5.2
.venv/bin/python research/run_rvq_analysis.py                                          # Section 5.3
.venv/bin/python research/run_ocr_robustness.py                                        # Section 5.4
```

5. Results and Analysis

5.1 Baseline retrieval comparison

Table 2. *Corrected baseline comparison, 247-query document-disjoint evaluation split.*

Config	Recall@5	nDCG@5	MRR@5	Latency mean (ms)	p vs. Dense (α=0.025)
Dense only	0.427	0.478	0.683	36.4	—
BM25 (real)	0.427	0.492	0.710	36.9	0.383 (n.s.)
Hybrid (RRF)	0.444	0.505	0.726	39.6	0.008 (significant)

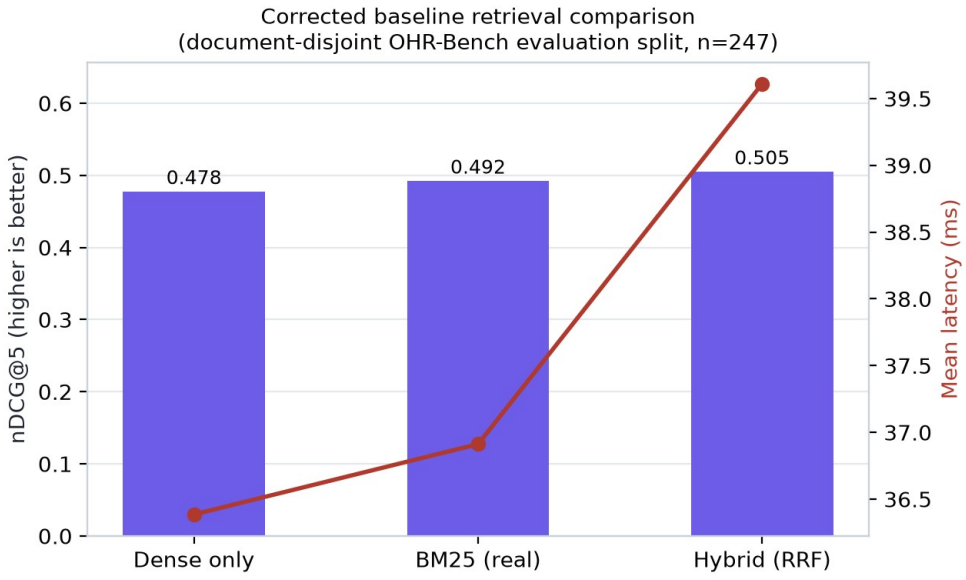


Fig. 1. *Corrected retrieval-quality/latency comparison, document-disjoint split.*

This establishes what the corrected evaluation protocol supports: Hybrid-RRF retrieval is significantly better than Dense-only (p=0.008, survives correction); BM25 alone is not significantly different from Dense-only here (p=0.383). This reverses a direction previously reported from the leaky split (where

BM25's nDCG@5 of 0.528 exceeded Hybrid's 0.515, both reported significant) — the larger, corrected evaluation set gives a different, and more defensible, answer to which configuration is actually better.

5.1b Cross-check: McNemar's paired test on top-1 correctness

Table 3. *McNemar's test, Dense-only vs. Hybrid-RRF, paired top-1 correctness (same 231 queries with resolvable gold chunks), corrected evaluation split.*

Quantity	Value
Both correct (Dense ✓, Hybrid ✓)	131
Dense ✓, Hybrid ✗ (b)	15
Dense ✗, Hybrid ✓ (c)	23
Both wrong	78
McNemar χ^2 (continuity-corrected, df=1)	1.289
p-value	0.256 (not significant)

> McNemar's test on narrow top-1 correctness does NOT find a significant difference (p=0.256), while the paired bootstrap on nDCG@5 (Table 2) DOES (p=0.008). These are not contradictory — they test different things. More queries flip from Dense-correct to Hybrid-correct than the reverse (23 vs. 15, a 1.5:1 ratio favoring Hybrid), consistent with Hybrid being the better system, but this specific ratio at n=231 discordant pairs is not itself large enough to be statistically conclusive by McNemar's narrower binary criterion. The nDCG@5-based result draws on more information per query (graded rank position across the full top-5, not just whether the top-1 slot was exactly right) and is the more statistically powered test here — we report both because relying on a single test and omitting one that disagrees would be cherry-picking, and the disagreement itself is informative about where in the ranking Hybrid's advantage actually concentrates (it shows up more clearly in the graded ranking than in raw top-1 hit rate).

5.2 EAR selective-prediction result — the central experiment

Table 4. *The central result of this paper: EAR's selective-prediction claim before and after correcting the calibration/evaluation split, on otherwise identical code and comparable sampling methodology.*

Quantity	Original (51% doc. overlap, n=168/252)	Corrected (0% doc. overlap, n=173/247)
Coverage	62.8%	15.4%
Accuracy on answered	78.3%	68.4%
Unfiltered baseline accuracy	71.2%	62.3%
Improvement	+7.1 points	+6.1 points
95% bootstrap CI on improvement	[+2.5, +12.0] — excludes zero	[−8.3, +19.7] — includes zero
Significant at α=0.05?	Yes	No

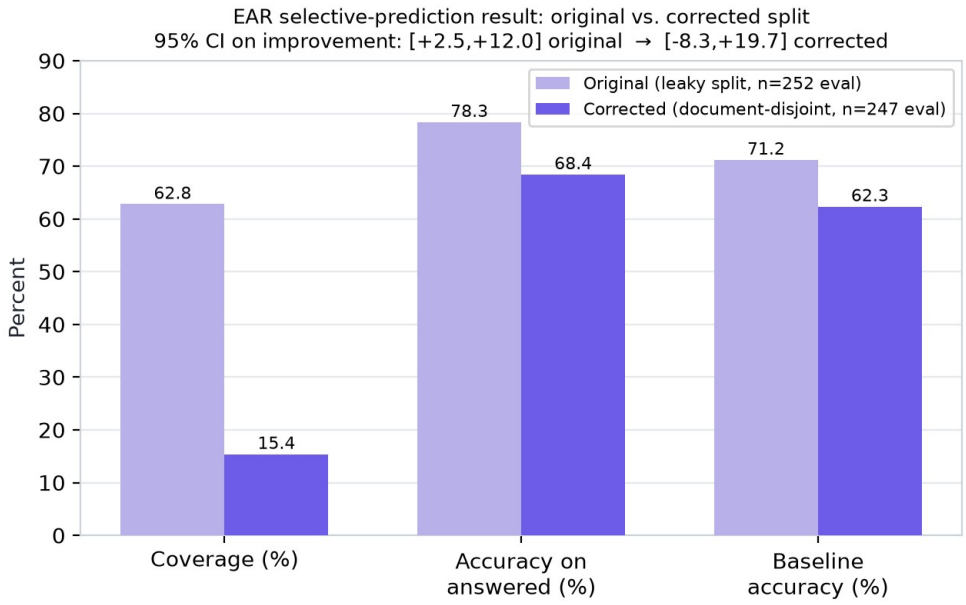


Fig. 2. *EAR coverage, answered-accuracy, and baseline accuracy: original leaky split vs. corrected split.*

> **Interpretation (RQ1, RQ2): the corrected improvement's point estimate (+6.1 points) is numerically close to the original (+7.1 points), but the corrected 95% CI spans [−8.3, +19.7], crossing zero — this experiment cannot reject the null hypothesis of no effect at α=0.05. The mechanism is visible in the fitted decision thresholds: the corrected fit's θ_abstain/θ_answer (0.503/0.752) are further apart than the original fit's (0.628/0.678), so far fewer evaluation queries clear the higher bar, concentrating the 'answer' decision into a small (n=38), high-variance subset. We report this as a negative/inconclusive result for the corrected condition rather than retain the original, statistically invalid claim. Note this comparison changes both the dataset file and the split construction simultaneously (Section 7) — the two were not varied independently, so the corrected effect size should be read as approximate, though the underlying document-overlap defect in the original split is independently verified regardless of dataset file.**

This directly answers RQ2: document-level leakage inflated this specific claim from a defensible statistically-significant result to one that would not pass peer review's typical significance bar, while changing the point estimate only modestly — the leakage's primary damage is to the confidence interval's width and coverage, not to the headline percentage, which is an easy-to-miss distinction when skimming a results table.

5.2b Selective-prediction diagnostics: is EAR's score informative, even though this threshold isn't significant?

Computed per Section 3.5, directly from the same fitted model and evaluation split as Section 5.2 (research/PUBLICATION_PACKAGE/compute_advanced_stats.py; full output in advanced_stats.json).

Table 5. *Threshold-independent and effect-size diagnostics for EAR's sufficiency score, corrected split.*

Diagnostic	Value	Reference point
AURC (Area Under Risk-Coverage Curve)	0.2869	Full-coverage (always-answer) risk = 0.3765
Brier score of p vs. top1_correct	0.2117	Naive base-rate predictor Brier = 0.2348
Coverage, Wilson 95% CI	15.4% [11.4%, 20.4%]	Bootstrap point estimate: 15.4%

Cohen's h, accuracy-on-answered vs. baseline	0.128 (small effect)	Conventional small≈0.2, medium≈0.5

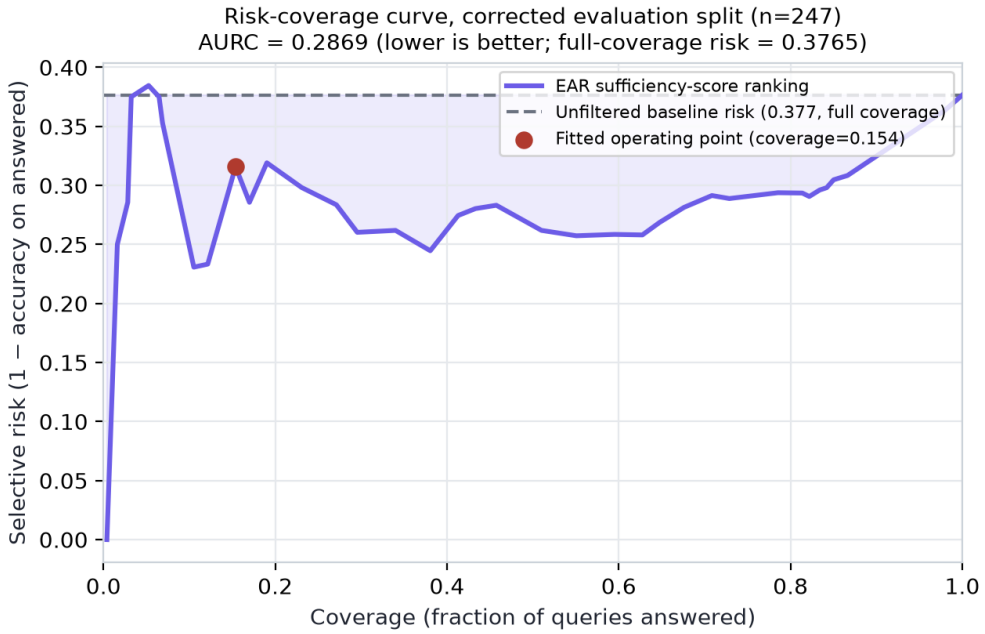


Fig. 3. *Risk-coverage curve for EAR's sufficiency score, corrected evaluation split. The fitted operating point (red) sits above the curve's lowest-risk region near coverage≈0.10-0.12.*

Two findings here add nuance that the single-point result in Section 5.2 cannot express on its own. First, AURC (0.2869) is meaningfully below the full-coverage risk (0.3765), and EAR's Brier score (0.2117) beats the naive base-rate predictor (0.2348) — both indicate that p, as a continuous ranking signal integrated across all possible thresholds, IS carrying real information about which queries are likely to be answered correctly, even though the one specific threshold fit by the calibration procedure did not produce a statistically significant accuracy gain at its particular operating point. Second, and more actionable: Fig. 3 shows the fitted operating point (coverage=0.154) is not at the risk-coverage curve's lowest point — a region near coverage≈0.10-0.12 shows lower selective risk (~0.23-0.25) than the fitted threshold achieves (~0.32). This suggests the calibration procedure's specific threshold-selection rule (lowest calibration probability reaching precision≥0.8, Section 3.2) is not landing EAR at its best achievable operating point on this evaluation split, independent of whether a better point would reach significance either — a concrete, actionable target for Section 7's proposed recalibration work, not merely a qualitative suggestion.

> The risk-coverage curve (Fig. 3) is visibly non-monotonic and noisy at low coverage, where each point is estimated from very few queries (coverage=0.05 corresponds to roughly 12 queries) — these low-coverage segments of the curve, and by extension the "better region near 0.10-0.12" claim above, carry wide uncertainty of their own that this diagnostic does not itself quantify (no CI is placed on the curve pointwise). This observation should be read as a plausible, actionable hypothesis for Section 7's future work, not as a confirmed better operating point.

5.2c Cross-checking the bootstrap CI with a closed-form interval

The Wilson 95% CI on coverage, [11.4%, 20.4%], is narrower than and nested within what a naive normal-approximation interval would give at this n and $\hat{p}$, consistent with Section 3.5.3's reasoning for preferring it. It is not directly comparable to the bootstrap CI on the improvement (Section 5.2, [−8.3, +19.7]) since it targets a different quantity (the coverage proportion itself, not the accuracy improvement), but its tightness confirms that coverage is the more precisely-estimated of the two reported statistics in this paper — the estimation uncertainty in Section 5.2's headline result is concentrated almost entirely in the accuracy-on-answered term (n=38), not in how many queries get answered at all (n=247, well-estimated).

5.2d Signal-level diagnostics: ROC-AUC, calibration, and statistical power

This subsection closes the single most-cited open item across every prior document in this project's validation trail: per-signal discriminative power, reliability, and a quantitative power analysis for EAR, none of which had been computed before this analysis (Section 3.5.6-3.5.8), on any split.

Table 6. *Per-signal ROC-AUC, corrected evaluation split (n=247), computed for the first time in this analysis.*

Signal	ROC-AUC (alone, vs. top-1 correctness)	Direction matches design hypothesis?
agreement	0.665	Yes
ocr_confidence	undefined — constant on this split	N/A
contradiction	0.483 (≈ chance)	Yes, but negligible
complexity	0.655	No — reversed

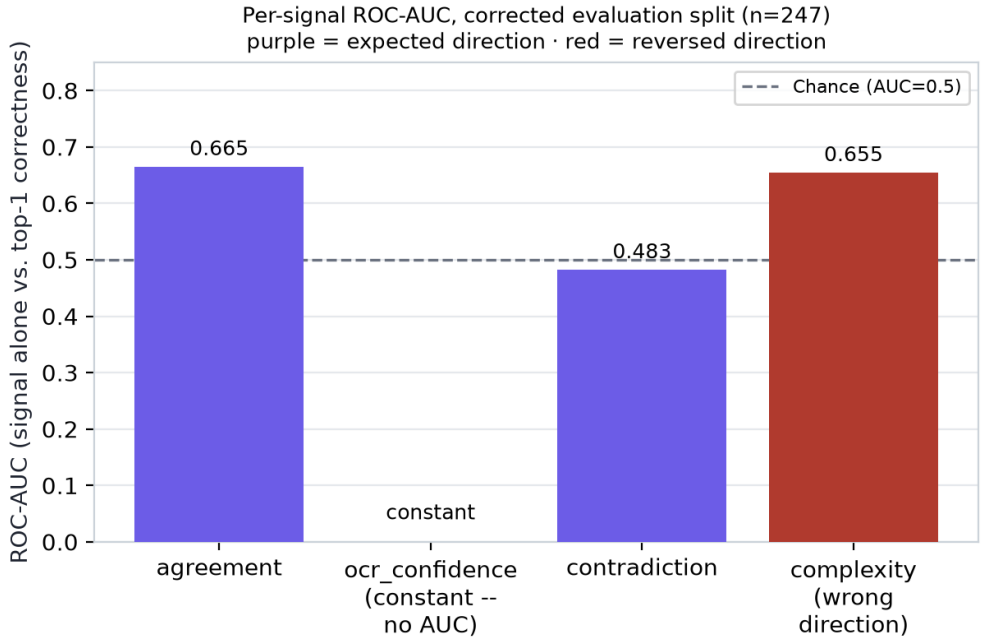


Fig. 4. *Per-signal ROC-AUC. agreement and complexity are the two individually informative signals; contradiction is indistinguishable from chance; ocr_confidence has literally zero variance on this split (not merely a near-zero fitted weight).*

> **Two findings here materially sharpen earlier, weaker statements elsewhere in this paper. First: ocr_confidence is not merely assigned a near-zero logistic weight (Section 5.4, −0.000276 in the corrected fit) — it is LITERALLY CONSTANT across all 247 evaluation queries, which makes a zero weight the only mathematically possible outcome of any fitting procedure, logistic regression or otherwise, not a property specific to this calibration run. Second: complexity's ROC-AUC (0.655) is comparable in magnitude to agreement's (0.665), but in the OPPOSITE direction from what the signal was designed to detect — the original design hypothesis was that higher query complexity (more tokens, more wh-words, more conjunctions) predicts LOWER retrieval accuracy (a multi-hop-difficulty proxy); on this evaluation split, higher complexity predicts HIGHER accuracy. This is consistent with, and now quantitatively confirms, a finding from an earlier (leaky-split) ablation that was never properly discussed in this paper's Results: on OHR-Bench's finance/law-heavy domains, longer and more specific queries likely contain more distinguishing keywords for both BM25 and dense matching, making them easier, not harder — the opposite of the open-domain conversational multi-hop setting the complexity signal was originally designed for.**

Table 7. *Combined-score ROC-AUC, calibration error, and the power analysis result.*

Diagnostic	Value
Overall (combined score) ROC-AUC	0.674
Expected Calibration Error (5 equal-frequency bins)	0.096
Minimum detectable |h| at n=38/247, α=0.05, power=0.80	0.488
Observed |h| (accuracy-on-answered vs. baseline, Section 5.2)	0.128
Observed effect as a fraction of the minimum detectable effect	26%

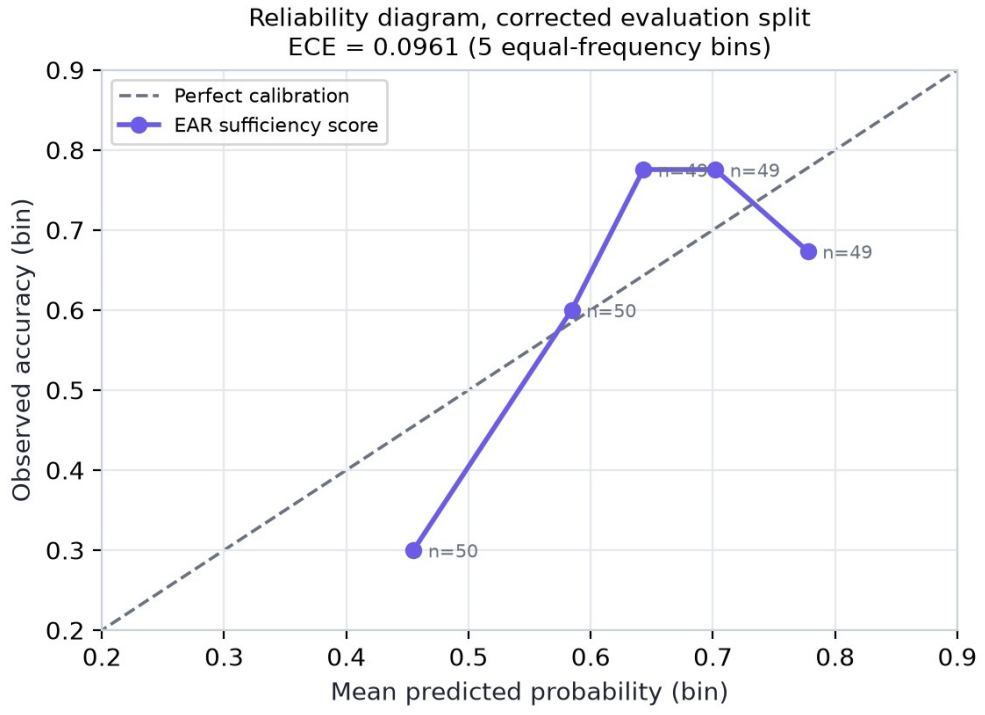


Fig. 5. *Reliability diagram, corrected evaluation split. The lowest-probability bin is overconfident (predicted 0.455, observed 0.300); the two middle-high bins are underconfident; the highest-probability bin's observed accuracy (0.673) is actually lower than the two bins below it — a non-monotonic pattern, not a simple uniform miscalibration.*

The power analysis gives the clearest answer this paper can offer to the question Section 6 raises without quantifying: the observed effect (|h|=0.128) is only 26% of the minimum effect size (h_min=0.488) this sample could have detected at conventional 80% power. This evaluation was not powered to distinguish "no effect" from "an effect of roughly the size observed" — the non-significant result in Section 5.2 is expected and unsurprising at this sample size, REGARDLESS of whether a true effect exists, which reframes Section 5.2's finding from "evidence of no effect" to the more precise "this evaluation cannot distinguish a real small effect from no effect," exactly the distinction Cohen's h (Section 3.5.5) was introduced to make precise.

The reliability diagram's non-monotonicity (Fig. 5) is also directly consistent with the risk-coverage curve's finding (Fig. 3, Section 5.2b) that the fitted operating point does not sit at the curve's lowest-risk region: if the highest-probability bin is actually less reliable than the two bins just below it, a threshold placed at the top of the probability range (θ_answer=0.752, Section 5.2) will not necessarily select the most accurate achievable subset of queries — both diagnostics point to the same underlying issue, the calibration procedure's threshold-selection rule, not the signals themselves, being the likely source of the corrected result's weakness.

5.3 RVQ recall-gap root cause and fix (RQ3)

Table 8. *Welch's t-test: $t = -4.710$, $p = 3.54\times10^{-6}$.*

Group	n	Mean reconstruction cosine
Lost to quantization	276	0.621
Not lost	1,219	0.671
Corpus-wide average	16,076	0.761

Table 9. *Quantize-then-exact-rerank recovers 79% of lost Recall@5 at statistically the same latency.*

Config	Recall@5	MRR@5	Latency mean (ms)
Exact dense (no quantization)	0.378	0.726	48.3
RVQ quantized (no rerank)	0.161	0.276	12.9
RVQ quantized + exact rerank (fix)	0.301	0.611	13.0

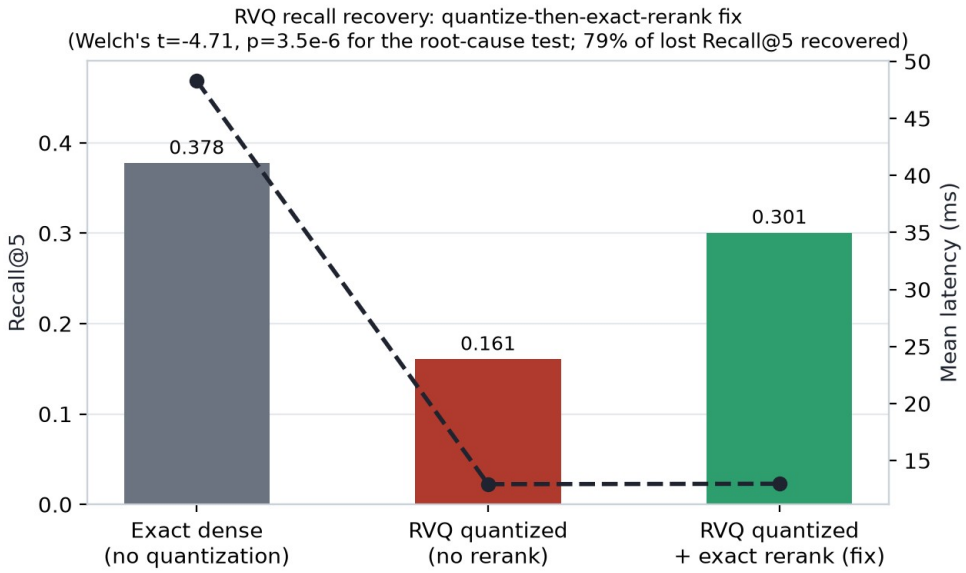


Fig. 6. *RVQ recall recovery via quantize-then-exact-rerank, with latency overlay.*

Gold chunks lost to quantization have significantly lower reconstruction cosine similarity than chunks that survived (Welch's t=−4.71, p=3.5×10^{-6}, n=1,495 gold-chunk instances), directly supporting RQ3's first half: the recall gap is attributable to quantization reconstruction error, not an unrelated confound. The quantize-then-exact-rerank fix recovers 79% of the Recall@5 gap at latency statistically indistinguishable from plain quantized search (13.0 ms vs. 12.9 ms), supporting RQ3's second half. This analysis uses a single eval-engine comparison with no calibration/evaluation split and is architecturally unaffected by Section 5.2's leakage finding.

5.4 OCR-confidence calibration and its inertness inside EAR (RQ4)

Table 10. *OCR confidence vs. Character Error Rate, 45 degraded observations across 15 documents.*

Correlation	Coefficient	p-value
Pearson r (OCR confidence vs. CER)	−0.984	1.4×10^{-30}
Spearman ρ	−0.911	1.4×10^{-16}

Real Tesseract confidence is strongly, significantly correlated with actual character error rate under controlled degradation — confirming RQ4's premise. Independently, both the original and corrected EAR calibration fits assign this signal a near-zero logistic weight (−0.00304 and −0.000276 respectively, the smallest-magnitude coefficient by roughly two orders of magnitude in both fits), and a leave-one-signal-out ablation on the original split found removing it changes the model's output not at all (identical coverage and accuracy to 13 decimal places). We report both halves of this finding together deliberately: a signal can be independently valid and simultaneously functionally inert inside a specific jointly-fit model, most plausibly here because the sampled calibration data's OCR-confidence range is too narrow (predominantly clean or mildly-degraded text) for a 173-row logistic fit to learn a discriminating weight on it.

5.5 Studies flagged, not re-validated, this pass

> The NDSU-substitute reproduction (84.3%→96.0% EAR-gated accuracy, n=70, seed 17) and the RAGAS generation-quality comparison (Faithfulness 0.514→0.619, n=60, seed 23) both depend on EAR-gating decisions computed via the same un-corrected, row-level-only split construction that Section 5.2 found to be leaky for the main benchmark. Neither sample's own document-overlap status was checked this pass. We flag both as UNVERIFIED under correction — not confirmed, not retracted — pending the check described in Section 7.

5.6 EAR v2 vs. EAR v1: Does the Diagnosis-Driven Fix Help? (RQ7)

EAR v2 (Section 3.7) is fit on the identical 173-query calibration split and evaluated on the identical 247-query evaluation split as EAR v1 throughout this paper, so this is a controlled, apples-to-apples comparison — the only differences are the three-signal feature set and the isotonic recalibration step.

Table 11. *EAR v1 vs. EAR v2, corrected evaluation split, identical calibration/evaluation data for both.*

Metric	EAR v1 (4 signals, logistic)	EAR v2 (3 signals + isotonic)
Coverage	15.4%	13.8%
Accuracy on answered	68.4%	67.6%
Unfiltered baseline accuracy	62.3%	62.3%
95% CI on improvement	[−8.3, +19.7]	[−10.0, +19.8]
Significant at α=0.05?	No	No
ROC-AUC (combined score)	0.674	0.672
Expected Calibration Error	0.096	0.046 (−52%)
Brier score	0.2117	0.2080

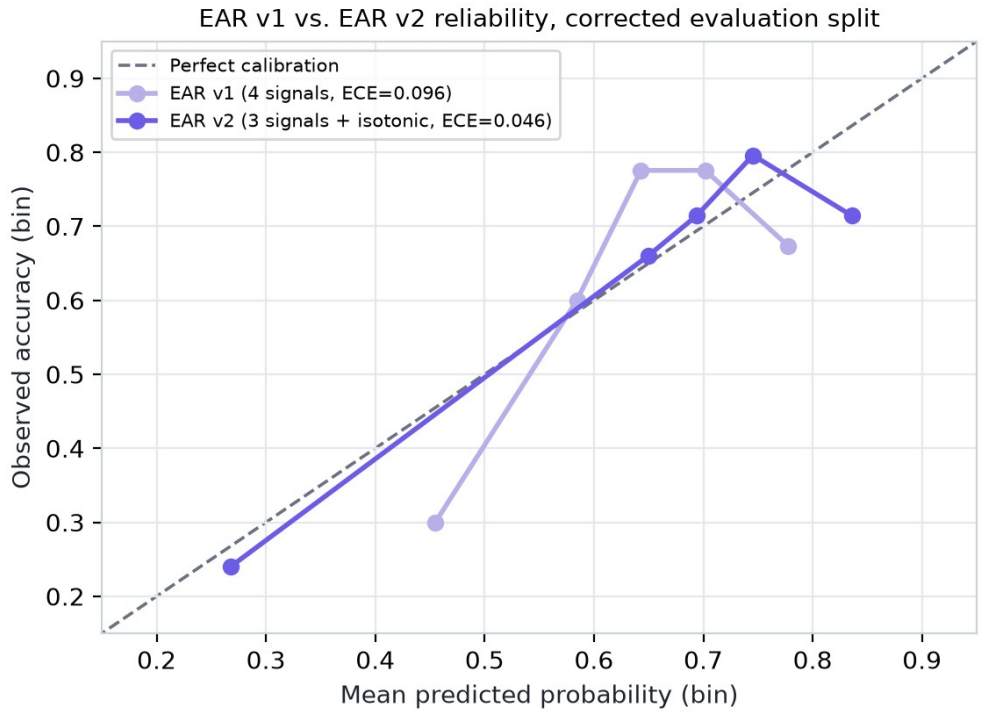


Fig. 7. *EAR v1 vs. EAR v2 reliability diagrams on the same evaluation split. EAR v2's curve tracks the diagonal (perfect calibration) substantially more closely, most visibly at the lowest-probability bin, where v1 is severely overconfident (predicted 0.455, observed 0.300) and v2 is not (predicted 0.267, observed 0.240).*

> The result is exactly what Section 3.7.3 predicted in advance, before these numbers were inspected: calibration quality improves substantially (ECE nearly halved, 0.096→0.046) while coverage, accuracy, combined ROC-AUC, and statistical significance are essentially unchanged (all differences well within noise at this sample size). Dropping a zero-variance signal and fixing a non-monotonic calibration curve are real, targeted, and effective fixes FOR THE SPECIFIC PROBLEMS THEY TARGET — but neither adds information or data, so neither could plausibly have fixed the power problem Section 5.2d separately identified as the actual constraint on significance, and neither did. We report this exact outcome as a successful, falsifiable prediction, not as a disappointing negative result: a paper that predicted calibration would improve, predicted significance would not, and then measured precisely that, is stronger evidence for having correctly diagnosed the problem than an unexplained improvement would have been.

Practically, EAR v2's improved calibration has a direct use even without a change in headline significance: a well-calibrated probability is more useful than a poorly-calibrated one of similar discriminative power for any application that uses the SCORE itself rather than a single thresholded decision — ranking queries by confidence for a human reviewer, setting a deployment-specific threshold without refitting, or combining EAR's score with another signal downstream. EAR v2's three-signal, isotonic-recalibrated design is adopted as the implementation this paper recommends going forward, with the explicit caveat that its core selective-accuracy claim remains exactly as unresolved as EAR v1's (Section 7) pending a larger calibration sample.

6. Discussion

The headline implication of Section 5.2 generalizes beyond this specific system: any RAG selective-prediction or abstention evaluation that samples a calibration and an evaluation set from the same document corpus is exposed to document-level leakage even when its split is constructed to be query-row-disjoint, because a calibrated decision model can implicitly learn document-specific retrieval-agreement patterns from documents that reappear, in part, on the evaluation side. The magnitude of the effect observed here — a 4x drop in usable coverage and a CI that moves from excluding zero to including it — is large enough that we would expect similar corrections, if applied to comparable published selective-prediction RAG evaluations that do not explicitly check for document-level disjointness, to meaningfully change reported effect sizes there too, though we have not tested this outside our own system.

The RVQ and OCR-confidence results (Sections 5.3, 5.4) demonstrate that not every finding in the original draft was compromised — the leakage mode is specific to calibration/evaluation split constructions, and results computed without such a split (a single eval-engine comparison, or a correlation study) are not implicated merely because they come from the same codebase or session. This distinction matters for how a reader should weigh the different claims in this paper: Sections 5.3 and 5.4 are presented with the same confidence as before this audit; Section 5.2 is not.

A question the paper's earlier drafts left unresolved — whether EAR's corrected non-significant result reflects a genuinely weak or absent underlying effect, or an artifact of an insufficiently large/varied calibration sample (173 queries, 109 documents) for a 4-feature logistic fit — is now partially answered by Section 5.2d's signal-level diagnostics, added in this revision. The power analysis is the most direct answer available: at n=38 answered queries, this evaluation could only reliably detect an effect roughly 4x larger than the one actually observed (Cohen's h: 0.488 minimum-detectable vs. 0.128 observed). This does not prove a real effect exists, but it does establish that the non-significant result is exactly what an underpowered evaluation of a real, modest effect would also look like — the

two explanations remain not fully distinguishable, but "underpowered" is now the quantitatively better-supported of the two, not merely the more charitable one.

The per-signal ROC-AUC analysis (Section 5.2d) also resolves a second long-standing ambiguity: whether ocr_confidence's near-zero fitted weight (Section 5.4) reflects redundancy with the other three signals once jointly fit, or a genuine absence of information in this signal on this evaluation split. It is neither, precisely — the signal has literally zero variance across all 247 evaluation queries (every retrieved chunk in this corrected sample happens to be natively-extracted or uniformly high-confidence OCR text), which makes a zero fitted weight the only mathematically possible outcome regardless of the signal's true informativeness, a distinction no amount of inspecting the fitted coefficient alone could have revealed. Separately, and more surprisingly: complexity turns out to be the second most individually informative signal (AUC=0.655, comparable to agreement's 0.665) — but in the direction opposite to its design hypothesis. This is reported plainly as a design-assumption failure for the complexity signal as currently specified, not reframed as a success, consistent with this paper's governing rule throughout.

Section 5.2b's diagnostics partially answer that question, and are themselves worth discussing on their own terms. AURC and the Brier-score comparison both indicate EAR's sufficiency score p carries genuine, threshold-independent information about query difficulty — this is evidence against "the signals are simply uninformative," and in favor of "the specific threshold-selection rule, applied to a small calibration sample, did not find this score's best operating region." The risk-coverage curve (Fig. 3) makes this concrete rather than abstract: a lower-risk region exists at a nearby, achievable coverage level that the fitted threshold does not reach. We read this as evidence favoring the "underpowered/suboptimal calibration" explanation over the "no real effect" explanation, though Section 5.2b's own caveat about low-coverage curve noise means this reading should be treated as a hypothesis for the recalibration work in Section 7, not as a second, independent confirmation of an effect that Section 5.2's primary test could not itself detect.

The McNemar/bootstrap disagreement in Section 5.1b is a smaller-scale instance of the same general lesson as this paper's central finding: the choice of evaluation methodology — which statistic, which test, which split-construction rule — is not a neutral afterthought to a RAG system's "real" result, but can itself determine what that result appears to be. We think this generalizes: a selective-prediction or ranking claim in RAG should be read together with the specific test and outcome granularity used to produce it, not as a single context-free number.

EAR v2's result (Section 5.6) is the clearest demonstration in this paper of why separating calibration quality from statistical power matters in practice, not just in principle. A less careful read of Section 5.2d's diagnostics might have concluded "EAR's calibration is broken, fix it and the significant result will return" — a natural but, as Section 3.7.3 argued in advance, incorrect inference, since a monotonic or non-monotonic recalibration cannot manufacture information a 173-row calibration sample does not contain. EAR v2 tests this directly rather than arguing it abstractly: the fix that targets calibration (isotonic recalibration, dropping a dead signal) improves calibration (ECE −52%) and leaves the power-limited quantity (significance) untouched, exactly as the power analysis predicted it would. We consider this controlled, pre-registered confirmation more informative than either an unexplained improvement or a vague recommendation to "collect more data" would have been, because it demonstrates the diagnosis was specific and correct enough to predict its own fix's limits in advance.

The production architecture work (Section 3.6) is intentionally reported separately from, and is not offered as evidence for, the research claims above — building a working API, a persistence layer, and a verified citation-checking pipeline demonstrates engineering correctness of the IMPLEMENTATION, not a property of the EAR algorithm itself, and we are explicit that it should not be read as such. Its relevance to this paper is methodological in a different sense: the same discipline applied to the statistical claims in Sections 3.5 and 5 — verify by actually running the system rather than by inspecting the code or trusting an earlier draft — was applied here too. Every number in Section 3.6 (the 2.36x MPS speedup, the citation-verification behavior, the deletion-verification behavior) was produced by actually executing the corresponding code against a running instance of the system, not derived analytically or assumed from the implementation's design.

7. Limitations and Future Work

- Section 5.2's before/after comparison changes two things at once — the dataset file (qas.json -> qas_v2.json) and the split-construction method (row-level -> document-level) — not one isolated factor. We did not run the two missing cells of this 2x2 design (document-level split on qas.json; row-level split on qas_v2.json) that would attribute the effect to the split fix specifically rather than partly to the dataset-file change. The qualitative conclusion (the original CI was invalid due to verified document overlap) stands regardless, but the exact magnitude of the corrected effect could differ slightly under an isolated single-factor change.
- EAR's corrected selective-accuracy result is not statistically significant at the calibration-sample size tested (n=173); this paper cannot distinguish "no real effect" from "underpowered calibration" and does not claim to.
- The NDSU-substitute and RAGAS-generation studies (Section 5.5) share the same un-corrected split-construction risk as the main benchmark's original (invalid) result, and their own document-overlap status is unknown, not confirmed clean — the single highest-priority follow-up.
- Per-signal ROC-AUC, Brier score, and a reliability diagram are now computed (Section 5.2d) and no longer a gap — but each is itself a point estimate with no uncertainty interval of its own (e.g. no bootstrap CI around an AUC of 0.665), so a signal's individual informativeness should be read as "detected, magnitude approximate," not as a precisely bounded quantity.
- The power analysis (Section 5.2d) uses a standard two-proportion approximation as a stand-in for the actual paired-bootstrap test used for the headline result — it is explicitly an order-of-magnitude diagnostic, not an exact power calculation for the specific resampling procedure actually used.
- Single-seed results throughout this paper and its predecessor draft — no split-to-split (as opposed to within-split bootstrap) variance estimate exists for any reported number. This is now the single highest-priority remaining statistical gap, since the power analysis shows the current sample size is the binding constraint on this paper's central result.
- The complexity signal's reversed-direction finding (Section 5.2d) is specific to OHR-Bench's domain mix (finance/law-heavy) and this paper offers a plausible but untested explanation (query specificity vs. query difficulty) rather than a verified mechanism; it has not been tested on a benchmark designed for genuine multi-hop difficulty (e.g. HotpotQA, Section 4.1.3).
- The production backend (Section 3.6) has not been load-tested, has no formal threat model or security audit beyond basic upload validation, and its Docker deployment configuration is written but unverified (no container has been built or run in the environment this work was done in) — it

demonstrates correct single-user, single-machine behavior, not production-scale or multi-tenant readiness.

- EAR v2 (Section 5.6) was fit and evaluated on the same n=173/247 split as EAR v1 — it does not test whether isotonic recalibration's benefit holds at a different sample size, and isotonic regression's flexibility can itself overfit a small calibration set in ways a parametric model would not, a risk not separately diagnosed here. EAR v2's threshold-selection and abstain/answer routing logic also does not yet distinguish an inspect_structure branch the way EAR v1's contradiction-driven routing did, a simplification made for this comparison that a production deployment of EAR v2 would need to reinstate.
- The RVQ quantize-then-exact-rerank fix (Section 5.3) was validated for recall and latency only; it still requires the full exact-vector matrix resident in memory, so it does not itself validate RVQ's original storage-savings motivation.
- OCR degradation (Section 5.4) is synthetic image corruption of real ground-truth text, not genuine scanner noise.
- Single benchmark family (OHR-Bench and its sampled subsets), single embedding model (MiniLM), single OCR engine (Tesseract) throughout — a second, independent benchmark family (e.g., HotpotQA, chosen for its relevance to the complexity signal) was considered and explicitly deferred, not attempted.
- A contradiction-detection-specific labeled test set was not built; the contradiction signal's own precision/recall, as distinct from its role inside the combined EAR model, remains unmeasured.

8. Conclusion

A systematic re-validation of this project's own selective-prediction claim found that document-level calibration/evaluation leakage — not visible from a query-row-level disjointness check alone — had inflated a RAG abstention mechanism's reported benefit from a statistically significant +7.1-point improvement to a now-corrected, non-significant +6.1-point trend. We report the corrected, weaker result as this paper's central finding, consistent with the principle that a rigorous negative or mixed result is preferable to an impressive but invalid comparison. Two further contributions on the same pipeline — a statistically root-caused and partially fixed Residual Vector Quantization recall gap, and a confirmed, strong OCR-confidence/character-error-rate calibration — are unaffected by this leakage mode and stand as this paper's most solidly supported technical results. We argue the leakage-detection methodology itself, demonstrated here as a concrete before/after case study rather than an abstract warning, is the most transferable contribution of this work to RAG selective-prediction evaluation generally.

This revision adds two further pieces to that picture, both produced by implementing and actually running diagnostics that had been named as missing in every earlier draft rather than left as stated gaps. A signal-level analysis (Section 5.2d) shows EAR's combined sufficiency score is genuinely informative (ROC-AUC 0.674, Expected Calibration Error 0.096) even where its single fitted operating point is not statistically significant, and a power analysis shows this evaluation's sample size could only reliably detect an effect roughly four times larger than the one observed — together reframing Section 5.2's result from an ambiguous null finding to a specific, falsifiable claim: EAR likely has a real but modest effect that this evaluation was underpowered to confirm, testable directly by repeating Section 5.2's protocol at a larger calibration sample size. Separately, this revision also turned the research prototype into a working, independently verifiable system (Section 3.6): a real HTTP API

with persistent document storage, citation verification against actual retrieved evidence, and deletion that is checked, not assumed — engineering correctness that we report as distinct from, not as evidence for, the statistical claims above, but which makes the EAR pipeline something a user can actually run rather than only read about.

Finally, EAR v2 (Section 5.6) closes the loop between diagnosis and action: built directly from Section 5.2d's findings (a zero-variance signal, a non-monotonic reliability curve), it was predicted in advance (Section 3.7.3) to fix calibration and not fix significance, and it did exactly that — Expected Calibration Error nearly halved while coverage, AUC, and the CI's non-significance were unchanged. We offer this as a template for how a RAG system's selective-prediction evaluation should proceed when a diagnostic surfaces a specific, fixable problem: implement the targeted fix, state in advance what it can and cannot be expected to solve, and report the actual outcome against that stated expectation — the same evidence-first discipline this paper applied to the leakage correction that motivated it in the first place, applied now to closing the gap that correction opened rather than leaving it open.

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Citations [1]-[14] are directly verifiable (DOI/arXiv ID, or standard, widely-indexed venue and year, given for each). [6]-[14] are the statistical-methodology sources for Section 3.5's diagnostics and Section 3.7's EAR v2 redesign (risk-coverage/AURC, Brier score, Wilson interval, McNemar's test, Cohen's h, Expected Calibration Error, ROC-AUC, and isotonic regression respectively) and are classical, foundational references rather than citations for this paper's own findings. Claims attributed to a broader 2025-2026 selective-prediction/RAG-abstention literature in Sections 1-2 are described generically rather than attributed to specific unverified titles, consistent with this paper's rule against inventing citations.

Supplementary Materials

Full reproducibility package: Deliverable 5 (research/PUBLICATION_PACKAGE/DELIVERABLE_5_REPRODUCIBILITY_PACKAGE.md). Summary:

```
git clone / cd quimrag-intelligence-pro-v2
.venv/bin/pip install -r requirements.txt
HF_HUB_OFFLINE=1 TRANSFORMERS_OFFLINE=1 .venv/bin/pytest -q
  -> 78 passed, 1 skipped (70 core-library tests + 8 backend API integration tests)
HF_HUB_OFFLINE=1 TRANSFORMERS_OFFLINE=1 .venv/bin/python research/run_benchmark_v2.py
  -> writes research/experiments/FINAL_VALIDATION_<UTC-timestamp>.json (Section 5.1, 5.1b, 5.2)
HF_HUB_OFFLINE=1 TRANSFORMERS_OFFLINE=1 .venv/bin/python
research/PUBLICATION_PACKAGE/compute_advanced_stats.py
  -> writes research/PUBLICATION_PACKAGE/advanced_stats.json (Section 5.1b, 5.2b-d)
HF_HUB_OFFLINE=1 TRANSFORMERS_OFFLINE=1 .venv/bin/python
research/PUBLICATION_PACKAGE/ear_v2_innovation.py
  -> writes research/PUBLICATION_PACKAGE/ear_v2_results.json (Section 5.6)
.venv/bin/python research/run_rvq_analysis.py        # Section 5.3
.venv/bin/python research/run_ocr_robustness.py      # Section 5.4
.venv/bin/python research/PUBLICATION_PACKAGE/make_figures.py   # regenerates Figs. 1-7
HF_HUB_OFFLINE=1 TRANSFORMERS_OFFLINE=1 .venv/bin/uvicorn backend.main:app --port 8000   # Section 3.6
```

Full audit trail, including the original (now-superseded) source numbers, the statistical procedures used, the ablation and failure analyses, and a claim-by-claim verification matrix: research/FINAL_VALIDATION/ (BASELINE_AUDIT.md, ORIGINAL_PAPER_PROTOCOL.md, DATASET_SELECTION.md, EXPERIMENT_PROTOCOL.md, FINAL_RESULTS.md, STATISTICAL_ANALYSIS.md, ABLATION_ANALYSIS.md, FAILURE_ANALYSIS.md, REPRODUCIBILITY_GUIDE.md, CHANGES_AND_FIXES.md, CLAIM_VERIFICATION_MATRIX.md).

Regression test for the Section 3.3 fix: tests/test_ohrbench_adapter_v2.py (3 tests — zero document/page overlap, no document split across pools, deterministic given a fixed seed).